Ajay Venkatesh

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Education

New York University, New York, NY — *Aug 2024 – May 2026*

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Computer Architecture, Data Structures

Experience

- **Campus Mesh** — *Jan 2026 – Present*
  AI Software Engineer — New York, US
  - Built **CampusIQ**, an LLM-powered AI assistant, with iterative **prompt engineering** and **RAG retrieval** over structured data and **embedding-based semantic search**, improving contextual accuracy by **30%**.
  - Designed a personalized **recommendation engine** using collaborative filtering on user interaction signals, increasing engagement by **28%** and retention by **50%**.
  - Developed scalable **Node.js microservices** powering real-time academic workflows for **10K+ users** across study groups, messaging, notifications, and profile management.
- **Amazon** — *May 2025 – Aug 2025*
  Software Development Engineer Intern — Seattle, US
  - Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
  - Implemented a **Model Context Protocol (MCP)** server for efficient tool-augmented retrieval, cutting AWS Athena query time and compute costs by **40%**.
  - Built a high-throughput, event-driven incident triage system on **AWS ECS Fargate, SQS, SNS**, decoupling distributed message processing and reducing response latency by **50%**.
  - Provisioned resilient **Infrastructure-as-Code** using **AWS CDK** on auto-scaling ECS clusters, ensuring **99.9%** availability under 2x peak load.
- **New York University (Novel AI Technologies, Inc.)** — *Aug 2024 – Present*
  AI Software Engineer — New York, US
  - Architected an **NSF-funded ($100K)** AI-powered health platform for **iOS** and **Android** ingesting real-time wearable telemetry; implemented **time-series anomaly detection** using **Isolation Forest** and **LSTM** models.
  - Built end-to-end **computer vision pipelines** (**YOLOv8, EfficientNet**) for user-submitted food images, integrated with **GPT-4 vision workflows** for nutritional estimation and dietary feedback.
  - Developed a full-stack insurance analytics product (**React, Node.js, REST APIs**) integrating **LLM-based document analysis** (**GPT-4 + LangChain**) for underwriting insights.
  - Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with concurrency control; processed **250K+** records and powered **AWS QuickSight** dashboards.
- **PricewaterhouseCoopers (PwC)** — *Aug 2021 – Aug 2024*
  Senior Software Engineer — Bengaluru, India
  - Designed a distributed backend on **Microsoft Azure** integrating with **Microsoft Dynamics** and supporting **ML inference pipelines** for healthcare analytics.
  - Built an **NLP-powered OCR pipeline** using **LayoutLM transformer** and Azure Form Recognizer to automate document data extraction, reducing processing time by **75%**.
  - Optimized data ingestion using **parallel processing** on **100K+** records, reducing execution from **5 hours to 1 hour**.
  - Led code reviews and mentored engineers on **software design and test-driven development**, reducing production bugs by **25%**.

Projects

- **InviLite (AI-Driven Data Engineering & Analytics Platform)** | *AWS, SageMaker, Bedrock, FAISS, Python*
  - Architected a cloud-native data pipeline (Kinesis, Lambda, S3 Bronze/Silver/Gold, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
  - Built a **RAG system** (SVM classification + FAISS vector indexing + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%** via feature engineering and model retraining.
- **Hospital Management System** | *React, Node.js, Express, MongoDB, SageMaker, Hugging Face*
  - Built a full-stack healthcare platform (React, Node.js, Express, MongoDB) managing appointments, patient records, and lab reports with **role-based access control**.
  - Fine-tuned a **Hugging Face transformer-based model** reaching **88% accuracy** for health condition prediction and deployed on **AWS SageMaker** for real-time diagnostic support.

Skills

- **Programming Languages:** Python, Java, C++, JavaScript (ES6), TypeScript, SQL
- **AI/ML:** PyTorch, TensorFlow, LLM Fine-tuning, RAG (LangChain, FAISS), Transformers, NLP, Hugging Face, Computer Vision (YOLOv8, EfficientNet), Time-Series Modeling, MLflow, SVM, Random Forest, Pandas, NumPy
- **Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices
- **Databases:** MySQL, PostgreSQL, MongoDB, DynamoDB
- **Cloud & DevOps:** AWS (Lambda, SageMaker, Bedrock, ECS, S3, CloudWatch, CDK, API Gateway, IAM), Azure (AI Services, Function Apps), Docker, Kubernetes, CI/CD, Terraform
- **Tools & Platforms:** Git, Linux, Jupyter, Postman, Cursor, GitHub Copilot, Model Context Protocol (MCP), Life Sciences Research, Biomedical Research, Scientific Computing, Genomics, Neuroscience, Drug Discovery, Context Engineering, Agent Architectures, Evaluation Frameworks, Production Systems Development

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*

An Aquila-optimized SVM Classifier for Diabetes Prediction *(IEEE)*

Certifications

AWS Cloud Practitioner | Stanford Machine Learning (Coursera) | Azure Data Fundamentals